# Ruiyu Mao

+1 469 347 1975 | @ ruiyumao360@gmail.com | LinkedIn | GitHub | Richardson, TX, USA

## Research Interests

- **Computer Vision & Data-Efficient Learning:** active learning, submodular optimization, data selection, open-set learning, 3D object detection, BEV perception, LiDAR perception, LiDAR semantic segmentation, autonomous driving.
- **Efficient LLMs & Multimodal Large Language Models:** vision-language models, multimodal learning, vision token reduction, attention-driven self-compression, efficient training, efficient inference, learnability-aware training, active data selection, multimodal DPO.

## Reviewer Service

- **Reviewer:** CVPR, AAAI, NeurIPS, TMLR.

## Education

- **The University of Texas at Dallas** — Richardson, TX
  Ph.D. in Computer Science; **GPA: 3.62/4.00** — Jan. 2024 – May 2027 (Expected)
- **The University of Texas at Dallas** — Richardson, TX
  M.S. in Computer Science; **GPA: 3.62/4.00** — Aug. 2021 – Dec. 2023
- **Case Western Reserve University** — Cleveland, OH
  B.S. in Aerospace Engineering — Aug. 2017 – May 2021

## Publications

- **Inconsistency-Based Data-Centric Active Open-Set Annotation**. [**AAAI 2024**]
  - **Ruiyu Mao**, Xu Ouyang, Yunhui Guo.
  - Introduced NEAT, a data-centric active open-set annotation method that used CLIP features and label clusterability to detect known-class samples, and an inconsistency score between model predictions and local feature neighbors to select informative queries while avoiding unknown classes.
- **STONE: A Submodular Optimization Framework for Active 3D Object Detection**. [**NeurIPS 2024**]
  - **Ruiyu Mao**, Sarthak Kumar Maharana, Rishabh K. Iyer, Yunhui Guo.
  - Proposed STONE, the first submodular optimization framework for active 3D object detection, with a two-stage algorithm that first selected representative point clouds via gradient-based submodular subset selection and then greedily rebalanced label distribution across categories and difficulty levels, achieving strong label efficiency on KITTI and Waymo.
- **Vision Token Reduction via Attention-Driven Self-Compression for Efficient Multimodal Large Language Models**. [**BigData 2025**]
  - Omer Faruk Deniz, **Ruiyu Mao**, Ruochen Li, Yapeng Tian, Latifur Khan.
  - Designed an attention-driven self-compression mechanism that leveraged the LVLM's own attention patterns to aggregate dense vision tokens into a compact set of summary tokens, aiming to remove most redundant vision tokens while preserving accuracy and remaining compatible with optimized attention implementations.
- **Learnability-Driven Submodular Optimization for Active Roadside BEV Perception**. [**CVPR 2026**]
  - **Ruiyu Mao**, Baoming Zhang, Nicholas Ruozzi, Yunhui Guo.
  - Proposed LH3D, a learnability-driven submodular framework for roadside monocular 3D detection that explicitly modeled ambiguous roadside-only scenes and selected camera views that were both highly informative and reliably labelable on DAIR-V2X-I.
- **SELECT: A Submodular Approach for Active LiDAR Semantic Segmentation**. [arXiv] [Under review]
  - **Ruiyu Mao**, Sarthak Kumar Maharana, Xulong Tang, Yunhui Guo.
  - Developed SELECT, a voxel-centric submodular pipeline that first performed voxel-level submodular subset selection to find representative regions, then estimated voxel uncertainty with Monte Carlo dropout, and finally applied submodular maximization for point-level class balancing to improve LiDAR semantic segmentation on SemanticPOSS, SemanticKITTI, and nuScenes under limited annotation budgets.
- **Learnability-Aware Active Multimodal Direct Preference Optimization**. [In preparation]
  - **Ruiyu Mao**, Omer Faruk Deniz.
  - Developed an active multimodal direct preference optimization (DPO) framework that used learnability-aware scoring to select informative image–text preference pairs, improving multimodal alignment while significantly reducing preference data and training cost.